Bank Marketing Dataset – Exploratory Data Analysis (EDA)

This notebook contains a simple exploratory data analysis of the **Bank Marketing dataset** from the UCI Machine Learning Repository. The goal here is to understand the structure of the data, look at basic statistics, and visualize a few important patterns before doing anything more advanced.

I've kept the analysis straightforward and practical, focusing on what each step tells us about the dataset.

Importing Required Libraries

First, we import the libraries needed for data analysis and visualization. Pandas is used for handling the dataset, while Matplotlib and Seaborn are used for plotting graphs.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

Loading the Dataset

The dataset is loaded using `pandas.read_csv`. This file uses a semicolon (`;`) as the separator, which is why it is explicitly specified. Once loaded, the data is stored in a DataFrame for further analysis.

```
df = pd.read_csv("bank.csv", sep=";")
```

Previewing the Data

To get a quick idea of what the dataset looks like, we display the first five rows. This helps in understanding the columns, data types, and general structure of the data.

```
df.head()
```

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previ
0	30	unemployed	married	primary	no	1787	no	no	cellular	19	oct	79	1	-1	0
1	33	services	married	secondary	no	4789	yes	yes	cellular	11	may	220	1	339	4
2	35	management	single	tertiary	no	1350	yes	no	cellular	16	apr	185	1	330	1
3	30	management	married	tertiary	no	1476	yes	yes	unknown	3	jun	199	4	-1	0
4	59	blue-collar	married	secondary	no	0	yes	no	unknown	5	may	226	1	-1	0

Dataset Structure and Data Types

The `.info()` method is used to understand the structure of the dataset. It shows the total number of rows and columns, the data type of each column, and whether there are any missing values.

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4521 entries, 0 to 4520
Data columns (total 17 columns):
 #   Column     Non-Null Count  Dtype
```

```
 ---  ------     --------------  -----
 0   age        4521 non-null   int64
 1   job        4521 non-null   str  
 2   marital    4521 non-null   str  
 3   education  4521 non-null   str  
 4   default    4521 non-null   str  
 5   balance    4521 non-null   int64
 6   housing    4521 non-null   str  
 7   loan       4521 non-null   str  
 8   contact    4521 non-null   str  
 9   day        4521 non-null   int64
 10  month      4521 non-null   str  
 11  duration   4521 non-null   int64
 12  campaign   4521 non-null   int64
 13  pdays      4521 non-null   int64
 14  previous   4521 non-null   int64
 15  poutcome   4521 non-null   str  
 16  y          4521 non-null   str  
dtypes: int64(7), str(10)
memory usage: 600.6 KB
```

Descriptive Statistics (Numerical Columns)

Using `.describe()` gives summary statistics for numerical columns such as mean, standard deviation, minimum, and maximum values. This helps in understanding the distribution and spread of numerical features like age, balance, and duration.

```
df.describe()
```

```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	age	balance	day	duration	campaign	pdays	previous
count	4521.000000	4521.000000	4521.000000	4521.000000	4521.000000	4521.000000	4521.000000
mean	41.170095	1422.657819	15.915284	263.961292	2.793630	39.766645	0.542579
std	10.576211	3009.638142	8.247667	259.856633	3.109807	100.121124	1.693562
min	19.000000	-3313.000000	1.000000	4.000000	1.000000	-1.000000	0.000000
25%	33.000000	69.000000	9.000000	104.000000	1.000000	-1.000000	0.000000
50%	39.000000	444.000000	16.000000	185.000000	2.000000	-1.000000	0.000000
75%	49.000000	1480.000000	21.000000	329.000000	3.000000	-1.000000	0.000000
max	87.000000	71188.000000	31.000000	3025.000000	50.000000	871.000000	25.000000

Descriptive Statistics (Categorical Columns)

For categorical columns, `.describe(include="object")` is used. This shows information such as the number of unique values, the most frequent category, and its frequency.

```
df.describe(include="object")
```

```
/tmp/ipykernel_25190/702825166.py:1: Pandas4Warning: For backward compatibility, 'str' dtypes are included by
select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version.
Explicitly pass 'str' to `include` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on
how to write code that works with pandas 2 and 3.
  df.describe(include="object")
```

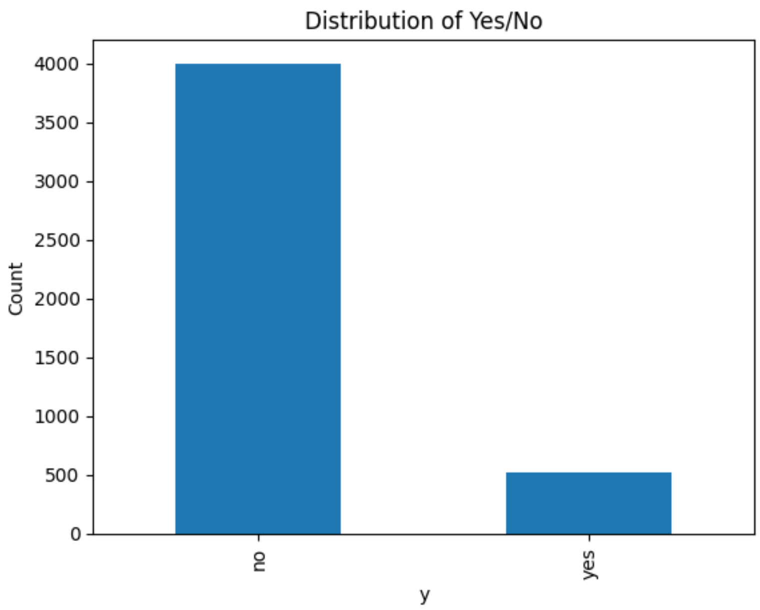
```
.dataframe tbody tr th {
    vertical-align: top;
}

.dataframe thead th {
    text-align: right;
}
```

	job	marital	education	default	housing	loan	contact	month	poutcome	y
count	4521	4521	4521	4521	4521	4521	4521	4521	4521	4521
unique	12	3	4	2	2	2	3	12	4	2
top	management	married	secondary	no	yes	no	cellular	may	unknown	no
freq	969	2797	2306	4445	2559	3830	2896	1398	3705	4000

Distribution of Target Variable (`y`)

The target variable `y` indicates whether a client subscribed to a term deposit. A bar chart is used to visualize how many clients said "yes" versus "no". This also helps identify class imbalance in the dataset.

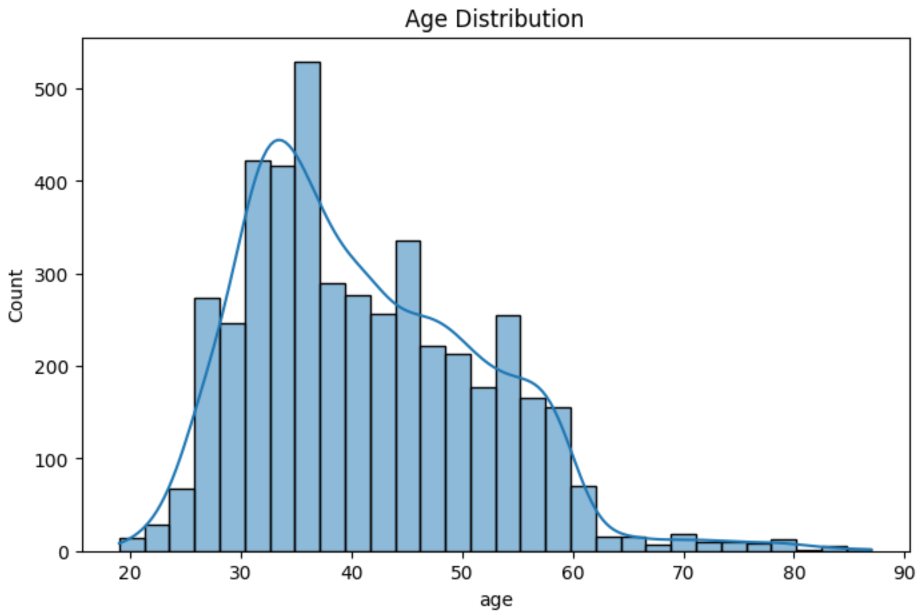
```
df["y"].value_counts().plot.bar()
plt.title("Distribution of Yes/No")
plt.xlabel("y")
plt.ylabel("Count")
plt.show()
```



Age Distribution

This histogram shows how client ages are distributed across the dataset. Most clients fall into the middle-age range, with fewer very young or very old clients.

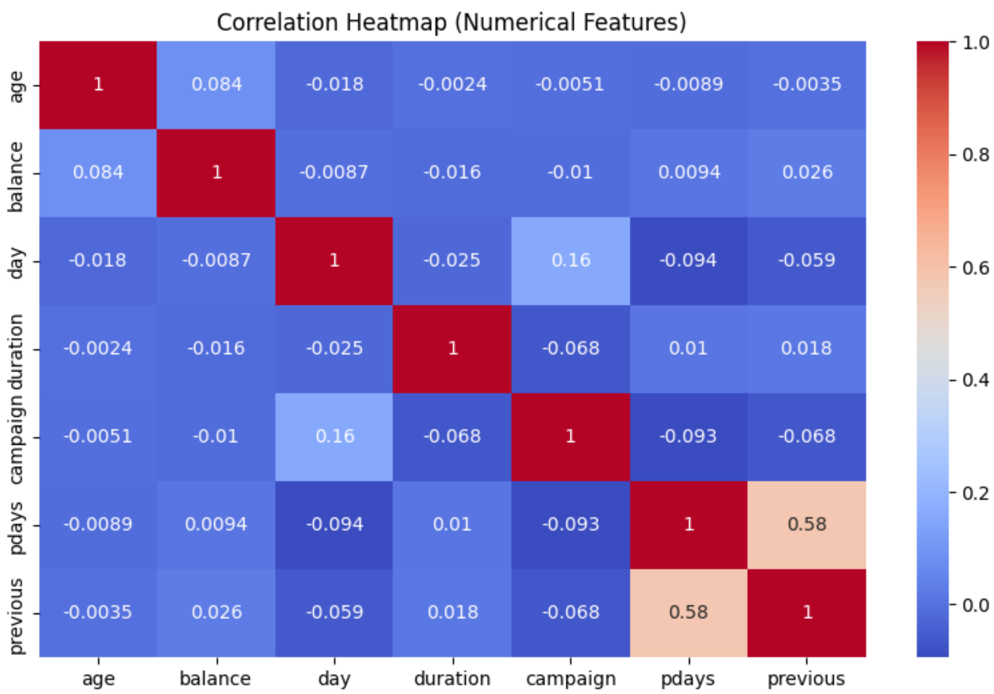
```
plt.figure(figsize=(8,5))
sns.histplot(df["age"], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```



Correlation Heatmap (Numerical Features)

Correlation analysis is performed only on numerical columns, since correlation is not defined for categorical data. The heatmap helps visualize relationships between numerical variables such as age, balance, duration, and campaign.

```
numeric_df = df.select_dtypes(include=["int64", "float64"])
corr_matrix = numeric_df.corr()
plt.figure(figsize=(10,6))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap (Numerical Features)")
plt.show()
```



Conclusion

In this analysis, the Bank Marketing dataset was explored using basic EDA techniques. The dataset contains both numerical and categorical features with no missing values. The target variable is imbalanced, with more clients not subscribing to the term deposit. Visualizations and summary statistics provide useful insights into customer characteristics and their potential influence on subscription behavior. This analysis can be used as a foundation for further modeling or prediction tasks.